SAMPLE IDENTIFICATION	
Code	Faro 12
Pit	1-GU-2-PA
Depth	1,738.00 m
Coordinates (well)	Lat. 1°20'54.75" S | Long. 51°25'32.84" W
Formation / Basin	Lighthouse / Amazon Basin
Age	Tournaisian–Visean (Lower Carboniferous)

GENERAL DESCRIPTION OF THE BLADE
Homogeneous clastic texture lamina. Grain-supported framework with dense packing. No regions with differential matrix concentration were identified; the rock is essentially clean (absence of significant clay matrix). No visible fractures. Primary intergranular porosity was strongly reduced by compaction and probable siliceous cementation. No significant open pores are observed. Discrete ferruginous films are observed sporadically at the intergranular contacts (more visible in the SW sector), without forming extensive concentrations. Larger grains (~0.30–0.40 mm) are homogeneously distributed, without granulometric bands or gradation.

Estimated modal composition (% vol)	
Monocrystalline quartz (Qz mono)	~70–75%
Polycrystalline quartz (Qz poly)	~5–8%
Feldspar (K-feldspar + plagioclase)	~5–8%
Lithic fragments	~3–5%
Micas (muscovite ± biotite)	~1–2%
Opaque minerals	~2–3%
Cement (siliceous + ferruginous)	~3–5%
Clay matrix	<2%
Visible porosity	<1%

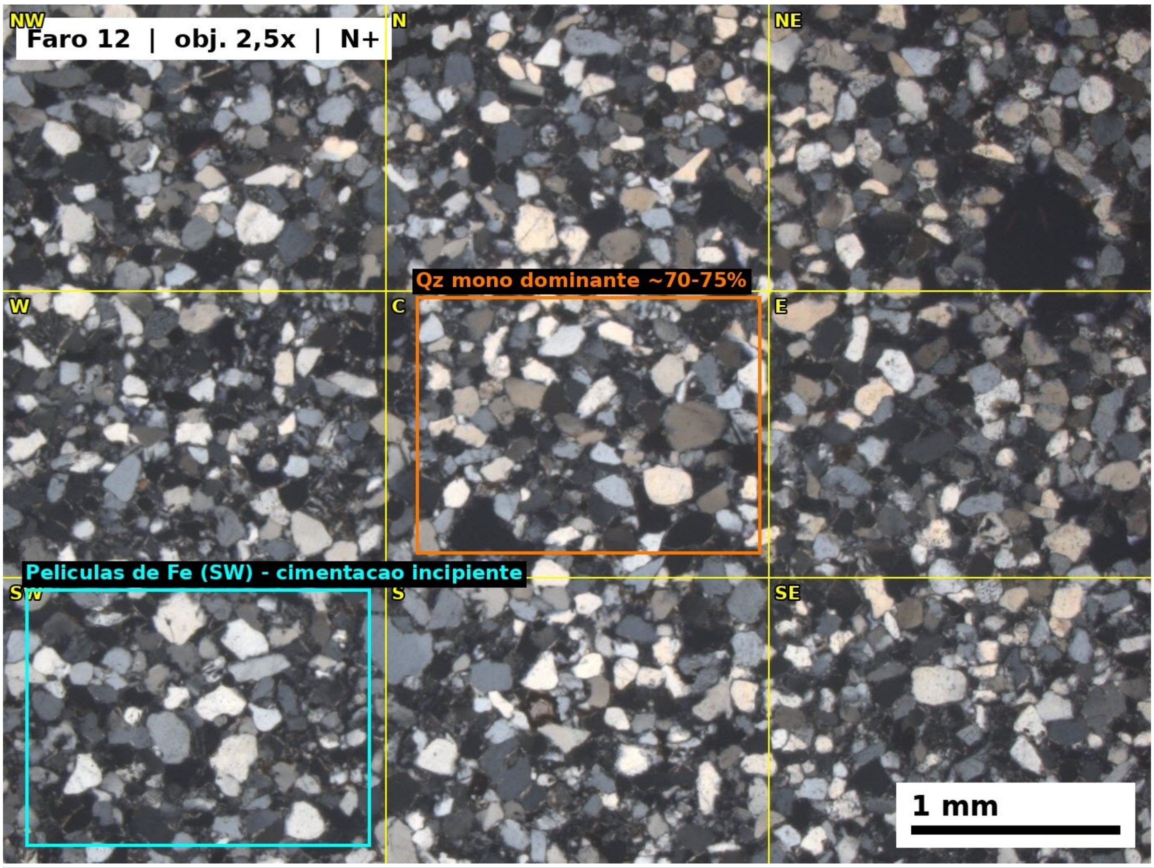
Composition:
Predominance of monocrystalline quartz with normal (non-undulating) extinction in clear grains, indicating a recycled high-grade plutonic/metamorphic source. Polycrystalline quartz with sutured internal contacts suggests a metamorphic source (basement quartzites). Rare feldspars, partially altered to clay minerals (eodiagenetic saussuritization).

Microscopic description	
Texture	Terrigenous clastic, grain-supported framework, homogeneous texture, without banding or gradation.
Particle size	Fine to medium sand (~0.12–0.35 mm). Estimated mode: fine sand (~0.15–0.20 mm).
Particle size selection	Moderately selected.
Packaging and contacts	Dense. Long (~50%), concave-convex (~30%), point (~15%), sutured (~5%). Compatible with advanced

	mechanical compaction and the beginning of chemical compaction.
Cement / diagenesis	Siliceous cementation (syntaxic growths/quartz overgrowth) inferred from the contacts between quartz grains and the difficulty in distinguishing original boundaries (a typical feature of mesodiagenesis in deeply buried quartzose sandstones). Intergranular iron oxide films (brownish hue) indicate eodiagenetic / telodiagenetic cementation by hematite/goethite. Sequence: mechanical compaction → clay infiltration + Fe-oxides (eodiagenesis) → chemical compaction + Qz overgrowth (mesodiagenesis) → oxidation (telodiagenesis).
Porosity	No evident open intergranular primary porosity, possibly eliminated by compaction + siliceous cementation.
Maturity	Textural: mature. Composition: mature to overripe (>70% Qz).

Sectoral Analysis of Grains					
Sector	**Mineral**	**Size (mm)**	**Sphericity**	**Rounding**	**Form**
NW	Qz mono	0.18	High	Subrounded	Equidimensional
NW	Qz mono	0.12	Average	Subangular	Equidimensional
NW	Qz mono	0.25	High	Subrounded	Equidimensional
N	Qz mono	0.30	Average	Subrounded	elongated
N	Feldspar	0.22	Average	Subangular	Prismatic
N	Qz poli	0.20	Low	Subangular	elongated
NE	Qz mono	0.15	High	Rounded	Equidimensional
NE	Opaque	0.10	High	Rounded	Equidimensional
NE	Qz mono	0.35	Average	Subrounded	Equidimensional
W	Qz mono	0.20	Average	Subrounded	Equidimensional
W	Lithic	0.28	Low	Subangular	elongated
W	Qz mono	0.14	High	Subrounded	Equidimensional
W	Qz mono	0.22	High	Subrounded	Equidimensional
W	Qz mono	0.16	Average	Subangular	Equidimensional
W	Feldspar	0.18	Average	Subangular	Prismatic
AND	Qz mono	0.25	High	Subrounded	Equidimensional
AND	Qz mono	0.12	Average	Subangular	Equidimensional
AND	Opaque	0.08	High	Rounded	Equidimensional
SW	Qz mono	0.20	Average	Subrounded	Equidimensional
SW	Qz poli	0.30	Low	Subangular	elongated
S	Qz mono	0.18	High	Subrounded	Equidimensional
S	Qz mono	0.15	Average	Subangular	Equidimensional

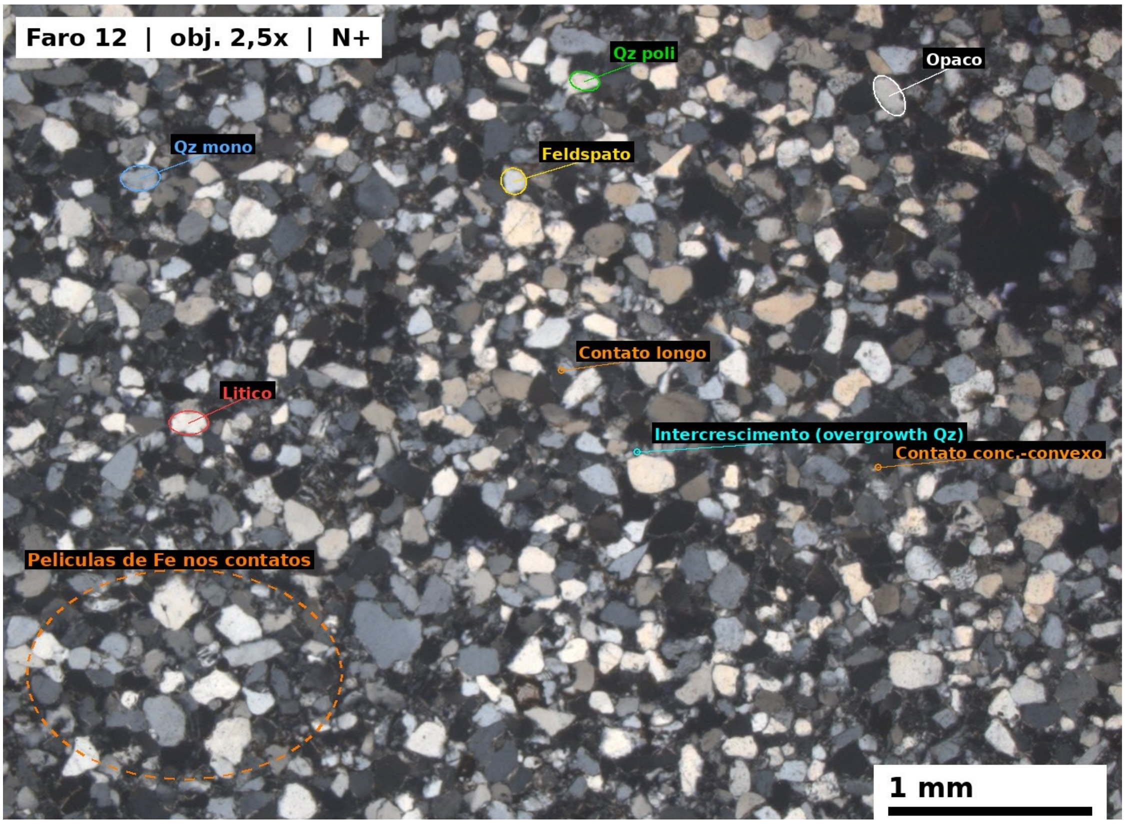
SE	Qz mono	0.22	High	Rounded	Equidimensional
SE	Lithic	0.20	Low	Subangular	elongated



Grain population: Dominant population: Monochromatic quartz , medium–high sphericity, subrounded, equidimensional, 0.15–0.22 mm. Subordinate population: Poly- and lithic quartz, low sphericity, subangular, elongated. Feldspars are rare and altered.

CLASSIFICATION
Fine to medium-fine quartz arenite, moderately sorted, mature.

PHOTOMICROGRAPHY (160 × 120 mm)



Photomicrograph (N+, 2.5× objective) of fine- to medium-grained, grain-supported, moderately sorted quartzarenite . Framework dominated by monocrystalline quartz (Qz mono), with subordinate polycrystalline quartz (Qz poly), feldspar (F), and lithic fragments (L). Predominantly long, concave-convex contacts, with intimate juxtaposition between quartz grains suggestive of siliceous cementation (overgrowth); discontinuous iron oxide films at the intergranular contacts indicate subordinate ferruginous cementation.